



GEEM 433

Engineers: Professionals for the Human Good

**ENGINEERING ETHICS AND MORALE
PHILOSOPHY**

Some Thoughts!

- A person's profession is a part of personal identity.
- Engineering is a profession.
- Engineering codes and guidelines from leaders in the field require engineers to work for the public good. This means improving people's well-being, welfare, and quality of life through their work.
 - *Promoting the well-being of the public*, not engaging in professionally prohibited actions, preventing harm to the public, and actively promoting the public's well-being.
 - *In designing for well-being*, engineers must keep in mind the social context of engineering and technology and the need for a critical attitude toward technology.

DRIVERLESS CARS!

It is easy to understand the advantages that driverless cars offer: reducing traffic collisions, providing better transportation for the elderly and disabled, lowering fuel consumption, and increasing traffic flow.

On the other hand, they raise many social, legal, and ethical questions. Perhaps the most obvious question is who should have responsibility for accidents.

The first fatal accident of a driverless car occurred in Williston, Florida, on May 7, 2016. A person in a Tesla driverless car was killed when a truck made a left turn in front of the car. The car went under the truck's trailer without braking, likely because neither the autopilot nor the driver saw the white side of the trailer against the bright sky.

DRIVERLESS CARS!

- Where should **moral responsibility** and **legal liability** lie in this case?

-> In this case, moral responsibility and legal liability are complex. Investigation revealed that the driver did not operate the Tesla according to the instructions and Tesla also did not have a system to check if the driver was ready to take over the control of the car when needed. There's also the question of how realistic it is to expect a driver to stay fully alert when an autopilot system is installed.

- Liability and responsibility are **not the only questions** raised by driverless cars.
 - How safe are they?
 - What kinds of information should be given to drivers before they purchase or use these vehicles?
 - How should the potential problems of hacking and terrorism be handled? (To handle the risks of hacking and terrorism with driverless cars, strong security measures are needed.)
 - What about the potential loss of driving-related jobs? Should there be retraining for other jobs?

Questions of Responsibility and Moral Issues

- Questions about **responsibility** aren't just related to driverless cars; they arise in engineering disasters like the **Challenger and Columbia space shuttle accidents**.
- There are important moral issues regarding how **technology** affects the **environment** and how **engineers** manage **risks**. These concerns matter to engineers not just because they design the technologies but also because being an engineer involves **professionalism**, which has a strong moral aspect.
- Engineers must think carefully about their **responsibilities** and the **impact** of their work on society and the environment.

The two components of professionalism are:

- (1) **expertise in a certain area (accounting, law, medicine, engineering, etc.) and**
- (2) **adherence to moral guidelines, usually laid out in a formal code of ethics.**

Failure in either of these areas means one is deficient as a professional.

Three Stages in the Development of Professional Identity

- **Independent Operator.** Professionalism is meeting fixed and clearly defined guidelines and expectations that are external to one's character.
- **Team-Oriented Idealist.** Rather than identifying professionalism with fixed rules and behaviors, professionalism is about meeting the expectations of other professionals, especially those who are admired.
- **Self-Defining Professional.** At this level, a person's values are fully mixed with their profession's values. Professionalism becomes part of who they are. This stage is often not fully achieved until mid-life.

What is a Profession?

- The use of profess and related terms in the Middle Ages was associated with the public profession of a way of life that carried strict **moral rules** (**religious** and ethical duties).
- By the late seventeenth century, the term "profession" moved away from its religious meaning and became more **secular**. It started to describe people having right **qualifications** and **skills** to offer valuable services to others, such as doctors, lawyers, or teachers.
- **Three approaches to professionalism are especially important in understanding the concept and can be useful in understanding professional identity:**
 - **Sociological Account**
 - **Social Contract Account**
 - **Morally Permissible Account - offered by philosopher Michael Davis.**

What is a Professionalism?

Sociological Account - Society gives professionals (like doctors, engineers, lawyers) special status because they serve the public with expert knowledge..

Social Contract Account - Professionals have an unspoken agreement with society: "We give you **trust**, **money**, and **respect** — in return, you behave responsibly and work for the public good."

- On the one hand, **professionals agree** to attain a high degree of professional expertise, offer good service to the public, and follow ethical rules in their work.
- On the other hand, the **public agrees** to let professionals enjoy above-average wages, have social respect and prestige, and enjoy a considerable degree of freedom to regulate themselves.
- The idea of such an agreement creates a strong sense of responsibility on a professionals or a developing professional

Morally Permissible - A profession is a group of individuals working in the same field who voluntarily choose to serve the public by pursuing a morally desirable goal, such as helping others or promoting the public good. Davis' definition emphasizes that a profession consists of more than one person and includes a public element, serving as a way for people to earn a living and occupy their working hours. People enter the profession willingly, and it focuses on achieving a morally beneficial goal, such as curing the sick or promoting the well-being of society.

Characteristics of a Profession

- **Extensive Training:** Requires a significant period of intellectual training, usually obtained at a college or university.
- **Specialized Knowledge:** Professionals possess knowledge and skills essential for the well-being of society.
- **Monopoly on Services:** Only qualified individuals can offer certain professional services, and they control how others can enter the field and get trained.
- **Workplace Autonomy:** Professionals take decisions based on their judgment, not just based on orders from others.
- **Ethical Standard:** A set of rules about right and wrong, usually written in code of ethics, to ensure public well-being.

Engineering as a Profession

Engineering is clearly a profession by all three accounts.

- First consider the Sociological Account.
- **Extensive Training:** Becoming an engineer requires high level of training in the college or university.
- **Specialized Knowledge:** Possessing knowledge and skills is vital for engineers for the well-being of society.
- **Monopoly on Services:** The engineering profession does not have complete control over the practice of engineering, because, in some countries, such as the United States, one does not have to be a registered professional engineer (PE) in order to practice engineering.
- **Workplace Autonomy:** Engineers often have more autonomy in the workplace than many non-professionals.
- **Ethical Standards:** Engineers have ethical codes that ensure public safety, sustainability, and ethical practices.

Engineering as a Profession

Engineering is clearly a profession by all three accounts.

Social Contract Account

- Engineers are highly skilled professionals who provide competent services. While engineers have ethical codes to follow, losing their Professional Engineer (PE) registration for unethical behavior doesn't necessarily stop them from working in their field. Unlike other professions where losing a license means you can't practice, many engineers aren't required to be licensed to continue their work.
- However, engineers still enjoy high wages and significant social status as part of the benefits of their profession.